

# Cost Analysis - Property RAG System

**Date:** October 2025 **Project:** Property Data RAG System **Purpose:** Simplyphi Recruitment Submission

## Executive Summary

This document provides a comprehensive cost analysis for the Property RAG System, covering development, deployment, scaling, and operational costs across different scenarios.

**Key Findings:** - **Current Setup:** \$0/month (100% free tier usage) - **Small Scale (1000 queries/day):** ~\$15-25/month - **Medium Scale (10,000 queries/day):** ~\$150-200/month - **Enterprise Scale (100,000 queries/day):** ~\$1,500-2,000/month

## 1. Current Implementation Costs

### 1.1 Development Stack (FREE)

| Component       | Technology                    | Cost | Notes                           |
|-----------------|-------------------------------|------|---------------------------------|
| Vector Database | ChromaDB (Local)              | \$0  | Self-hosted, persistent storage |
| Embedding Model | Sentence-Transformers         | \$0  | Runs locally, no API calls      |
| LLM             | Google Gemini Pro (Free Tier) | \$0  | 15 req/min, 1500 req/day limit  |
| Backend         | FastAPI                       | \$0  |                                 |

| Component     | Technology | Cost | Notes                   |
|---------------|------------|------|-------------------------|
|               |            |      | Open source framework   |
| Frontend      | Streamlit  | \$0  | Open source framework   |
| Hosting (Dev) | Localhost  | \$0  | Development environment |

**Total Development Cost:** \$0/month

### 1.2 Infrastructure Requirements

**Minimal Setup:** - RAM: 4GB minimum (8GB recommended) - Storage: 1GB for code + 500MB for vector DB - CPU: 2 cores minimum - Bandwidth: Minimal (local processing)

**Cost for Self-Hosted:** \$0 (using existing hardware)

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## 2. Production Deployment Costs

### 2.1 Small Scale Deployment (1,000 queries/day)

**Infrastructure:** | Component | Provider | Specifications | Cost/Month |  
| |-----|-----|-----|-----| | **Server** | Digital Ocean Droplet | 2 vCPU, 4GB RAM | \$24 | | **Alternative** | AWS Lightsail | 2 vCPU, 4GB RAM | \$20 | | **Alternative** | Heroku Basic | 512MB RAM | \$7 |

**Services:** | Service | Provider | Usage | Cost/Month |  
|-----|-----|-----|-----| | **LLM** | Google Gemini Pro | 1,000 queries/day | \$0 (free tier) | | **Alternative LLM** | OpenAI GPT-3.5 | 1,000 queries × \$0.001 | \$0.30 | | **Vector DB** | ChromaDB (self-hosted) | Included in server | \$0 | | **Alternative Vector DB** | Pinecone Starter | 100K vectors | \$0 (free tier) |

**Total Small Scale (Gemini):** \$20-24/month **Total Small Scale (OpenAI):** \$50-54/month

## 2.2 Medium Scale Deployment (10,000 queries/day)

**Infrastructure:** | Component | Provider | Specifications | Cost/Month  
| |-----|-----|-----|-----| | **Server** | AWS EC2  
t3.medium | 2 vCPU, 4GB RAM | \$30 | | **Load Balancer** | AWS ALB |  
Basic | \$16 | | **Storage** | AWS EBS | 50GB SSD | \$5 |

**Services:** | Service | Usage | Cost/Month | |-----|-----|-----| |  
**LLM (Gemini Pro)** | 10,000 queries/day × \$0.001 | \$300 | | **LLM (GPT-3.5 Turbo)** | 10,000 queries/day × \$0.001 | \$300 | |  
**Embeddings** | Self-hosted Sentence-Transformers | \$0 | | **Vector DB**  
| Pinecone Standard (1M vectors) | \$70 | | **Alternative** | Weaviate  
Cloud | 10M vectors | \$25 | | **Monitoring** | Datadog Basic | 5 hosts |  
\$15 |

### Scenarios:

**Option A - Gemini + Self-hosted ChromaDB:** - Infrastructure: \$51/month - LLM: \$300/month - **Total: ~\$351/month**

**Option B - Gemini + Managed Vector DB:** - Infrastructure: \$51/month - LLM: \$300/month - Vector DB: \$70/month - **Total: ~\$421/month**

**Option C - OpenAI + Pinecone:** - Infrastructure: \$51/month - LLM: \$300/month - Vector DB: \$70/month - **Total: ~\$421/month**

## 2.3 Enterprise Scale (100,000 queries/day)

**Infrastructure:** | Component | Specifications | Cost/Month |  
|-----|-----|-----| | **Kubernetes Cluster** | 3 nodes ×  
t3.large (2 vCPU, 8GB each) | \$150 | | **Load Balancer** | AWS ALB with  
SSL | \$25 | | **Storage** | 200GB SSD | \$20 | | **CDN** | CloudFlare Pro |  
\$20 | | **Monitoring & Logging** | Datadog Pro | \$50 |

**Services:** | Service | Usage | Cost/Month | |-----|-----|-----| |  
**LLM (Gemini Pro 1.5)** | 100K queries/day @ \$0.001/query | \$3,000 |

| **Alternative: OpenAI** | 100K queries/day @ \$0.001/query | \$3,000 |  
| **Alternative: Claude** | 100K queries/day @ \$0.002/query | \$6,000 |  
| **Vector DB (Pinecone)** | 10M vectors, Standard | \$100 | |  
**Alternative: Weaviate** | 100M vectors | \$200 | | **Caching (Redis)** |  
AWS ElastiCache | \$50 |

**Total Enterprise Scale:** - Infrastructure: \$265/month - Services  
(Gemini + Pinecone): \$3,150/month - **Total: ~\$3,415/month**

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## 3. Cost Optimization Strategies

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### 3.1 LLM Cost Reduction

**Strategy 1: Hybrid Approach** - Use Gemini free tier for first 1,500 queries/day - Fallback to paid tier only when exceeded - **Savings:** ~\$45/month for small scale

**Strategy 2: Query Caching** - Cache common queries and responses - Estimated cache hit rate: 30-40% - **Savings:** 30-40% reduction in LLM calls

**Strategy 3: Local LLM for Simple Queries** - Use Ollama (free, local) for simple queries - Use cloud LLM only for complex queries - **Savings:** 50-60% reduction in API costs

### 3.2 Infrastructure Optimization

**Strategy 1: Spot Instances** - Use AWS Spot instances (70% discount) - **Savings:** ~\$100/month for enterprise scale

**Strategy 2: Reserved Instances** - 1-year commitment: 30% discount - 3-year commitment: 50% discount - **Savings:** ~\$150/month for enterprise scale

**Strategy 3: Serverless Architecture** - AWS Lambda for API (pay per use) - Only pay when queries are processed - **Ideal for:** Variable/bursty traffic

### 3.3 Vector Database Optimization

**Strategy 1: Compression** - Use quantization for embeddings - Reduce vector dimensions (384 → 256) - **Savings:** 33% storage cost reduction

**Strategy 2: Tiered Storage** - Hot data: Fast SSD storage - Cold data: S3 archival - **Savings:** ~40% for historical data

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## 4. Scaling Cost Projections

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### 4.1 Cost Per Query Analysis

| Scale      | Queries/Day | Monthly Queries | Infrastructure | LLM     | Vector DB | Total/ Month | Cost/Query |
|------------|-------------|-----------------|----------------|---------|-----------|--------------|------------|
| Dev        | 100         | 3,000           | \$0            | \$0     | \$0       | \$0          | \$0.000    |
| Small      | 1,000       | 30,000          | \$24           | \$0     | \$0       | \$24         | \$0.0008   |
| Medium     | 10,000      | 300,000         | \$51           | \$300   | \$70      | \$421        | \$0.0014   |
| Large      | 50,000      | 1,500,000       | \$150          | \$1,500 | \$100     | \$1,750      | \$0.0012   |
| Enterprise | 100,000     | 3,000,000       | \$265          | \$3,000 | \$100     | \$3,365      | \$0.0011   |

**Key Insight:** Cost per query decreases with scale due to infrastructure efficiency.

### 4.2 Break-Even Analysis

**Free Tier Limits:** - Gemini: 1,500 queries/day free - After limit: \$0.001 per query

**When to Upgrade:** - **Stay Free:** < 1,500 queries/day - **Basic Paid:** 1,500 - 10,000 queries/day - **Managed Services:** > 10,000 queries/day

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## 5. Alternative Architectures & Costs

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### 5.1 Fully Serverless (AWS)

| Component | Service             | Cost Model                                               |
|-----------|---------------------|----------------------------------------------------------|
| Compute   | AWS Lambda          | \$0.20 per 1M requests +<br>\$0.0000166667 per GB-second |
| Vector DB | Pinecone Serverless | Pay per query (\$0.0001/query)                           |
| LLM       | Bedrock Claude      | \$0.003 per 1K input tokens                              |
| Storage   | S3                  | \$0.023 per GB                                           |

**Estimated Cost (10K queries/day):** - Lambda: ~\$15/month - Pinecone: ~\$30/month - LLM: ~\$300/month - **Total: ~\$345/month**

### 5.2 Fully Open Source (Self-Hosted)

| Component  | Technology            | Cost          |
|------------|-----------------------|---------------|
| Server     | Own hardware/VPS      | \$30-50/month |
| Vector DB  | ChromaDB/Milvus       | \$0           |
| Embeddings | Sentence-Transformers | \$0           |
| LLM        | Ollama (Llama 3 8B)   | \$0           |
| Monitoring | Prometheus + Grafana  | \$0           |

**Total: \$30-50/month (ANY scale)**

**Trade-offs:** - Lower quality LLM responses - Higher latency - More maintenance overhead - Full data privacy

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## 6. Cost Comparison: Cloud Providers

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### 6.1 Vector Database Comparison

| Provider              | Free Tier               | Paid Tier           | Notes                       |
|-----------------------|-------------------------|---------------------|-----------------------------|
| <b>Pinecone</b>       | 100K vectors            | \$70/month for 1M   | Managed, easy setup         |
| <b>Weaviate Cloud</b> | 100K vectors            | \$25/month for 10M  | Open source based           |
| <b>Qdrant Cloud</b>   | None                    | \$0.40 per GB/month | Cost-effective at scale     |
| <b>ChromaDB</b>       | Unlimited (self-hosted) | Free                | Requires own infrastructure |

### 6.2 LLM Comparison

| Provider             | Model            | Input Cost                      | Output Cost  | Context     |
|----------------------|------------------|---------------------------------|--------------|-------------|
| <b>Google Gemini</b> | Pro              | Free (limited) then \$0.0005/1K | \$0.0015/1K  | 30K tokens  |
| <b>OpenAI</b>        | GPT-3.5 Turbo    | \$0.0005/1K                     | \$0.0015/1K  | 16K tokens  |
| <b>Anthropic</b>     | Claude 3 Haiku   | \$0.00025/1K                    | \$0.00125/1K | 200K tokens |
| <b>Meta</b>          | Llama 3 (Ollama) | Free (self-hosted)              | Free         | 8K tokens   |

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# 7. ROI & Business Case

## 7.1 Value Proposition

**For 10,000 queries/day @ \$421/month:** - Cost per query: \$0.0014  
- If each query saves 5 minutes of manual search: 833 hours/month saved - At \$20/hour labor cost: **\$16,660 value generated - ROI: 3,855%**

## 7.2 Pricing Strategy for SaaS

**Suggested Pricing Tiers:**

| Tier         | Queries/Day | Cost to Serve | Suggested Price | Margin |
|--------------|-------------|---------------|-----------------|--------|
| Free         | 10          | \$0           | \$0             | -      |
| Starter      | 100         | \$0.10        | \$10/month      | 99%    |
| Professional | 1,000       | \$1           | \$50/month      | 98%    |
| Business     | 10,000      | \$14          | \$200/month     | 93%    |
| Enterprise   | Custom      | Variable      | Custom          | 80-90% |

# 8. Cost Monitoring & Alerts

## 8.1 Recommended Alerts

| Metric       | Threshold   | Action                     |
|--------------|-------------|----------------------------|
| LLM API Cost | > \$100/day | Switch to cached responses |
| Query Volume | > 10K/day   | Upgrade infrastructure     |
| Error Rate   | > 5%        | Investigate issues         |

| Metric        | Threshold    | Action             |
|---------------|--------------|--------------------|
| Response Time | > 3s average | Scale up resources |

## 8.2 Cost Tracking Tools

- **AWS Cost Explorer:** Track AWS spending
  - **Pinecone Dashboard:** Monitor vector DB usage
  - **Custom Analytics:** Track query patterns and optimize
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## 9. Recommendations

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### 9.1 For Current Scale (Development)

✅ **Use Current Stack:** - ChromaDB (self-hosted): \$0 - Gemini Free Tier: \$0 - Local hosting: \$0 - **Total: \$0/month**

### 9.2 For Production (1-10K queries/day)

✅ **Recommended:** - AWS Lightsail/DO Droplet: \$20-30/month - Gemini Pro (paid): \$300/month - Self-hosted ChromaDB: \$0 - **Total: ~\$320-330/month**

### 9.3 For Enterprise Scale

✅ **Recommended:** - Kubernetes cluster: \$150/month - Gemini Pro with caching: \$2,000/month (33% savings) - Weaviate Cloud: \$200/month - Redis caching: \$50/month - **Total: ~\$2,400/month**

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## 10. Conclusion

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The Property RAG System is designed to be **cost-effective at all scales**:

1. **Development:** \$0/month (100% free)
2. **Small Production:** ~\$25/month

3. **Medium Scale:** ~\$350-450/month

4. **Enterprise:** ~\$2,400-3,400/month

**Key Cost Drivers:** - LLM API calls (70-80% of costs) - Vector database (10-15% of costs) - Infrastructure (10-15% of costs)

**Optimization Potential:** - Query caching: 30-40% savings - Hybrid LLM approach: 50% savings - Spot instances: 30-50% savings - **Total potential savings: 60-70%**

**Final Verdict:** The system provides exceptional value with costs scaling linearly with usage while maintaining high performance and accuracy.

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